

Tokenization of Real-World Assets Using the Distributed Ledger Technology

Laurie Simon Hodrick¹

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Digital assets, including **cryptocurrencies** and **stablecoins**², are rapidly becoming mainstream financial assets. Cryptocurrencies, the decentralized digital form of money that uses encryption for security on a shared **distributed ledger (blockchain)**, emerged after publication of Satoshi Nakamoto's 2008 whitepaper "Bitcoin: A Peer-to-Peer Electronic Cash System." This document laid the foundational blueprint for Bitcoin and the entire modern cryptocurrency industry, which now has a global market capitalization of \$2.28 trillion. **Digital native** Bitcoin accounts for 55.9% (\$1.27 trillion), and stablecoins like Tether (USDT), which emerged in 2014, account for 13% (\$302 billion).³

Adoption of **tokenized real-world assets**, including U.S. Treasuries, corporate stocks, and corporate bonds, is not yet nearly as widespread as the digital natives and is minuscule relative to the global financial system, with tokenized real-world assets currently accounting for \$38.17 billion in aggregate.⁴ Nevertheless, the expected growth in adoption of these assets is accelerating, with McKinsey estimating the market could reach \$2 trillion by 2030,⁵ and Citi Institute estimating a 2030 base case of \$5.5 trillion (bear case \$2.7 trillion, bullish case \$8.2 trillion).⁶ In March 2026 the U.S. Securities and Exchange Commission (SEC) approved NASDAQ's proposal to allow certain stocks and exchange-

¹ Visiting Fellow, The Hoover Institution, Stanford University and A. Barton Hepburn Professor Emerita of Economics in the Faculty of Business, Columbia Business School, Columbia University. This paper is prepared for the Hoover Institution Commons Project. I acknowledge Lara Druyan, Darrell Duffie, Robert Hodrick, Chris Labosky, and Jeff Strnad for their helpful comments and suggestions.

² Words in bold are defined in the glossary.

³ CoinGecko, Crypto Market Cap Charts. While Bitcoin was originally envisioned as a version of electronic cash for online payments, it is now better understood as a digital commodity rather than a medium of exchange. The example of Bitcoin illustrates the evolving nature of many crypto assets and some of the resultant challenges in designing appropriate regulations for them. See the discussion in Section 4.

⁴ RWA.xyz, a registry for real-world assets. Aggregate includes \$10 billion in active yield strategies.

⁵ From Ripples to Waves: The Transformational Power of Tokenizing Assets. (2024, June 20). McKinsey & Company.

⁶ Tokenization 2030: Wall Street On-Chain. (2026, June). Citi Institute.

traded products to trade and settle in tokenized blockchain-based form. The New York Stock Exchange (NYSE) is developing a digital platform for 24/7 **on-chain** trading and instant settlement of tokenized securities. Tokenized real-world assets therefore deserve our focus and attention now, as we anticipate and actively seek to shape what this market can and will become.

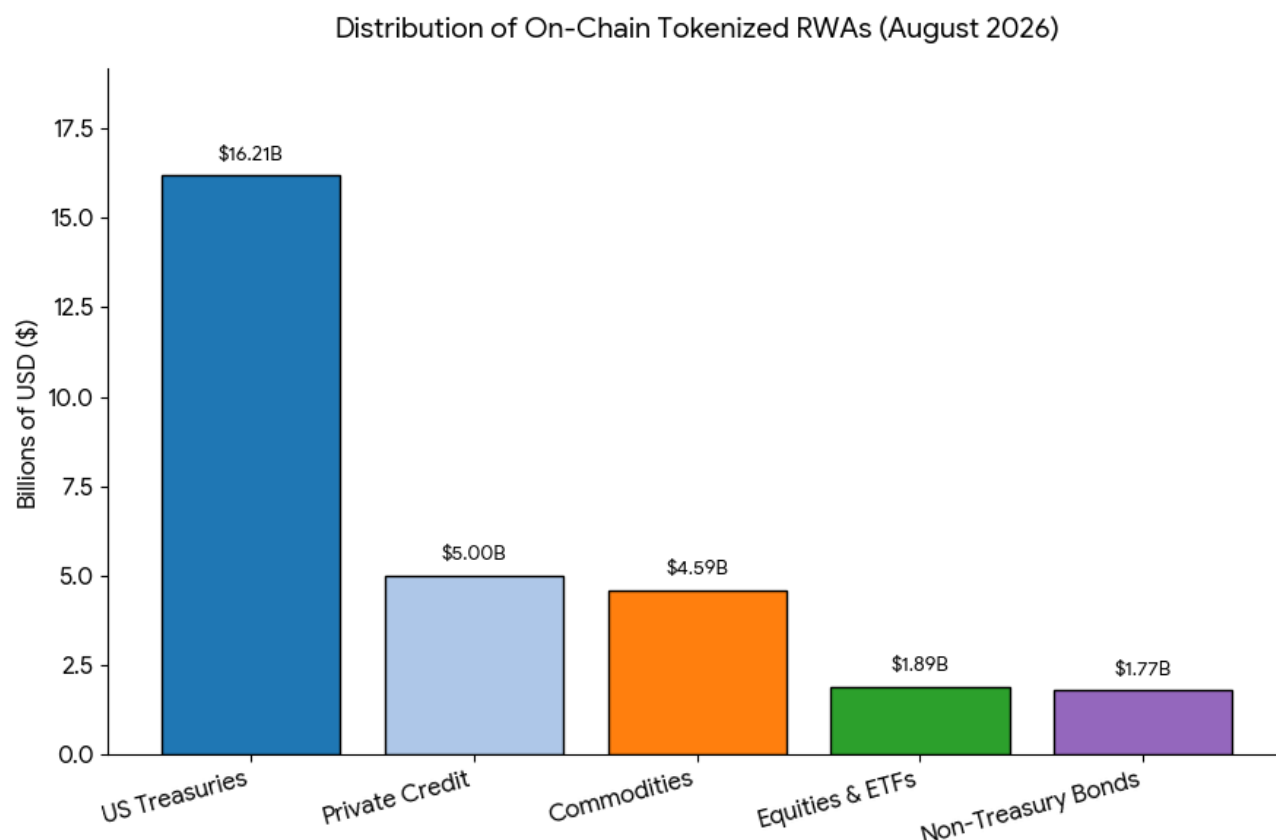
This paper begins with a working definition of tokenized real-world assets and emphasizes their implications for market transformation, including efficiencies, product innovation, and expanded market access. We focus on both the benefits and challenges of changing how assets are issued, traded, and settled, which is at the very core of how financial systems operate. This paper seeks to distinguish the economic benefits of tokenization from the legal, operational, and financial-stability risks that arise when on-chain representations are linked to **off-chain** assets.

The paper proceeds as follows. Section 1 defines tokenized real-world assets. Section 2 discusses their key benefits. Section 3 examines risks and challenges. Section 4 surveys current U.S. federal regulation. Because we are at the inception of these nascent assets, Section 5 concludes with guiding regulatory principles for policymakers.

Section 1. Tokenized Real-World Assets

Tokenized real-world assets are **smart-contract**-based assets issued on a shared distributed ledger (blockchain), that is, using cryptography for recordkeeping and security. The **token** provides a unique and **pseudonymous** digital representation, on a blockchain network, of an interest in a real-world asset, such as U.S. Treasuries, financial securities including corporate stocks and bonds, private credit, mutual funds and exchange-traded funds, real estate, art, or gold and other commodities. These digital representations can convey some or all of the legal and economic rights associated with ownership of the

underlying asset, for example, a token representing a share of equity could include all associated voting rights and dividend payments.⁷



Source: RWA.xzy

Many consider tokenized assets to be the next phase in the evolution toward a fully digitized economy, with security financing “likely to become the most important application of tokenized finance.”⁸ While early adoption of digitized assets has so far been concentrated in short-term U.S. Treasury debt, as demonstrated in the figure above,⁹ Larry Fink, CEO of BlackRock, predicted in his 2025 annual letter to investors that “the next step going forward will be tokenization of financial assets, and that means every stock, every

⁷ The scope of those rights depends on the tokenization model discussed in Section 4.

⁸ Tokenized Finance and the Perimeter of Central Banking. Darrell Duffie. (2026, August 22). Jackson Hole Symposium.

⁹ For further discussion of tokenizing Treasuries, see “Tokenizing Reserves and Treasuries,” Darrell Duffie, Chapter 7 in *Finishing the Job and New Challenges*, forthcoming, edited by Michael Bordo and John Cochrane, Monetary Policy Conference, Hoover Institution, Stanford University. (2025, May). Hoover Press.

bond...will be on one general ledger.”¹⁰ Growth is expected to be led by public market securities and highly liquid short-duration collateral assets, particularly U.S. Treasuries and equities,¹¹ with Fink anticipating that tokenized funds will become as commonplace as exchange-traded funds (ETFs), which are currently a \$22 trillion global market. As illustrative examples, asset managers are launching on the blockchain both digital money market funds, which are open-ended funds registered under the Investment Company Act of 1940, and private market funds, which are exempt from registration under the 1940 Act.¹²

Four elements are most often credited with the rapid growth of tokenized assets. First, there has been a decade of foundational infrastructure buildout and technological maturity, allowing the growth of on-chain money and settlement mentioned above. These building blocks, once at critical mass, will provide the embedded financial infrastructure needed for other tokenized assets to scale.

Second, large traditional financial market infrastructure providers, including the Depository Trust & Clearing Corporation (DTCC), NYSE, and NASDAQ, are now integrating tokenization into their core platforms with SEC participation. In July 2025, SEC Chairman Paul Atkins emphasized the goal of modernizing securities rules and regulations, thereby enabling America’s financial markets to fully embrace blockchain technology and move **on chain**, under “Project Crypto.” In December 2025, the SEC granted no-action relief for a DTCC three-year pilot offering a tokenization service for DTCC-custodied highly liquid assets including stocks, ETFs, and U.S. Treasuries, in partnership with almost 40 financial

¹⁰ CNBC, January 2024. Subsequently, BlackRock launched its tokenized USD Digital Liquidity Fund, BUIDL, on Ethereum for accredited institutional investors. BUIDL holds cash, U.S. Treasury bills and repurchase agreements, relying on standard SEC exceptions to avoid registration with the SEC. BUIDL now has \$2.5 billion in assets under management across multiple chains. WisdomTree Treasury Money Market Digital Fund (WTGXX) received SEC exemptive relief to trade and settle instantly, 24/7, as a registered mutual fund, in March 2026. Franklin Templeton’s registered funds secured SEC no-action letter approval in August 2026 to hold shares of Franklin OnChain U.S. Government Money Fund (FOBXX) for purposes of cash management and as collateral in securities lending programs. This excuses Franklin from having to replicate physical ledgers with **digital twin** blockchain-based representations of physical assets.

¹¹ Tokenization 2030, Wall Street On-Chain. (2026, June). Citi Institute.

¹² According to the Investment Company Institute (2026, August 26), money market funds held \$7.9 trillion in total assets, including \$3.1 trillion in retail money market funds and \$4.8 trillion in institutional money market funds.

firms and tech providers including JPMorgan Chase, Goldman Sachs, BlackRock, and Vanguard. Live trades have now been processed, with a full launch planned for October 2026. As of January 2026, the Intercontinental Exchange has developed an NYSE digital multi-chain platform for 24/7 trading and on-chain instant settlement of tokenized securities, for which it is currently seeking final, comprehensive SEC regulatory approval under Exchange Act Rule 19b-4. As of March 2026, NASDAQ received SEC approval to enable certain stocks and major ETFs, including those tracking the S&P 500 and NASDAQ 100, to be issued, traded, and settled in tokenized form. Also, in April 2026, the SEC approved an initial NYSE rule change that allows Russell 1000 constituents to trade in tokenized form during the DTC pilot. These movements towards a unified, instead of fragmented, infrastructure enhance legal certainty and investor protection, facilitating widescale deployment.

Third, there is a growing proportion of “digitally native” Generation-Z investors, born between 1996 and 2010, who view cryptocurrency as a core pillar of modern finance and demand digital assets. Along with younger millennials, most are using peer-to-peer (P2P) payment applications, and almost half own digital assets (which are four times more likely to be held than retirement accounts).¹³ They value speed and connectivity, trusting decentralized platforms and mobile apps that are “always on, always available, always-accessible” more than traditional legacy banking systems.¹⁴

Fourth, regulatory and legislative clarity over digital assets in the U.S. is improving and is thought to be accelerating the pace of adoption of tokenized assets, making digital assets more accessible without fear of enforcement action. In addition to the certainty offered by the SEC actions highlighted above, the SEC’s January 2026 Staff Statement on Tokenized Securities and the March 2026 SEC/ Commodity Futures Trading Commission (CFTC) joint interpretive framework provided additional guidance on the treatment of

¹³ What is Gen Z? (2024, August). McKinsey & Company. Note that while the tokenized real-world asset market to date is being used primarily by sophisticated financial institutions, retail demand will likely grow as these markets scale.

¹⁴ From Crypto to Climate: How Gen Z Can Redefine the Future of Cybersecurity. (2025, June 30). World Economic Forum.

tokenized securities and other digital assets, and the SEC’s recent proposed “Regulation Crypto Assets” in August 2026 may encourage capital formation through token offerings. The July 2025 Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act established a comprehensive federal framework for payment stablecoins. It provides consumer protection by, among other mechanisms, requiring full backing with cash or Treasuries. It also delivers payment rails for a broader tokenized asset ecosystem.¹⁵

Section 2. Key Benefits of Tokenized Assets

Tokens can be transferred, traded, and settled on the blockchain, providing the benefits of a digital asset. These include 24/7 global trading and near real-time settlement on a shared system of record. This enhanced liquidity and operational efficiency is further enhanced by programmability of smart contracts, allowing broader access to new types of asset classes and supporting product innovation.

Traditional markets operate for 32.5 hours a week with settlement on a T+1, trade date plus one business day, basis. In contrast, blockchains operate continuously, allowing potential 24/7 global trading with lower transaction costs and faster (near instant) settlement, mitigating counterparty exposure. Trade confirmation, clearing and settlement using traditional methods require multiple intermediaries and repeated reconciliation across independent networks.¹⁶ Each institution maintains its own records. A blockchain

¹⁵ This legislation has stimulated the rapid growth of tokenized money market funds. This may be, in part, because the legislation restricts stablecoin issuers from paying interest, making tokenized money market funds a yield-bearing on-chain alternative. It may also be because permitted stablecoin issuers can hold their required reserves in registered government money market funds under the GENIUS Act. For a fuller discussion of regulatory developments, see Section 4.

¹⁶ Equities have traditionally settled through stock exchanges and central securities depositories (the DTCC), money market funds have been settled internally by the mutual fund company or transfer agent, while corporate bonds have been processed and settled through the National Securities Clearing Corporation and the Depository Trust Company, managed by the DTCC. DTCC currently reports \$114 trillion in assets under custody, including \$74.1 trillion in equities, \$11 trillion in ETFs, and \$4.1 trillion in money market instruments. It processed and settled \$3.7 quadrillion in transactions in 2024. DTCC Launching Tokenization for \$144 Trillion Asset Market This July. (2026, May 4), *Global Finance*.

can consolidate fragmented clearinghouses and separate institutional ledgers, or complement a central authority, by providing a shared, transparent record.¹⁷

Tokens can furthermore be coded with specific rules (smart contracts) to ensure consistent execution.¹⁸ There should be fewer data errors since embedded compliance by design provides a higher degree of automation, such as computing interest calculations and coupon payments on corporate bonds, also reducing servicing costs. Other automations include the composability of transactions and contingent transactions triggered by time, price feeds, or the completion of another transaction.¹⁹ As all participants have access to the ledger, there is enhanced transparency, visibility, and auditability into transactions on chain.

Claims to assets can also be fractionalized, broadening the investor base for high-value assets by reducing the minimum investment, which is currently a barrier to entry. As servicing smaller investors becomes an economically viable proposition, the potential customer base and access to investments is expanded.²⁰ All of this creates the possibility of increasing liquidity and streamlining corporate action processes such as dividend or interest payments and proxy voting.

Tokenization is also fostering new products: For example, ETF issuers can tokenize their off-chain funds, with firms like Franklin Templeton and BlackRock already issuing tokenized funds to track assets on public blockchains. Given the possibility of near-instant settlement, an asset's yield can be computed and distributed intraday, which is a major shift from traditional finance, where yields, dividends, and interest are typically calculated and paid out on a monthly, quarterly, or annual basis due to legacy settlement delays (like

¹⁷ Continued modernization of existing financial market infrastructure may likely happen in parallel to the buildout of the blockchain infrastructure.

¹⁸ For example, FOBXX has embedded on-chain smart contracts to enforce Investment Company Act of 1940 compliance rules, including KYC/AML verification, programmatic dividend distribution, and transfer limitations (secondary market restrictions).

¹⁹ Darrell Duffie and Chaojun Wang (2026, June 21) develop an efficient contracting protocol in "Smart Contracts in Network Markets."

²⁰ Whether broader retail access to illiquid assets is desirable from an investor-protection perspective, however, is a separate question addressed in Section 3.

T+1), tokenized money market funds are paving the way for accrued dividends and interest to be reflected or distributed dynamically.²¹ Fractional claims can be traded on a range of previously less traded assets, including illiquid indivisible assets like works of art or other valued collectibles.

Section 3. Key Risks and Challenges of Tokenized Assets

The testing of tokenized assets to date has been on **permissioned ledgers**, to control security and compliance. In some cases, this is accomplished using a permissioned overlay on top of a permissionless underlying blockchain. Outside of these controlled environments, concerns arise about user privacy, security and cyber-security,²² technical feasibility, potential contagion of financial crises, and regulatory compliance.²³

Distributed ledgers can provide unprecedented transparency in financial markets but must preserve privacy protections for certain user information.²⁴ While cryptographic tools can help balance these concerns, it remains to be seen whether controlled, permissioned ledgers at scale are required to maintain privacy while ensuring Know Your Customer (KYC) and Anti-Money Laundering (AML) compliance.

Once tokens can trade on multiple platforms, the inefficiency of retrofitting and integrating legacy infrastructure and the lack of accepted interoperability standards may provide significant obstacles to the adoption and secondary market liquidity of digital

²¹ Traditional systems that calculate yield may need retooling to keep up with the speed of tokenized settlement cycles.

²² Security is a broad concept of keeping physical or digital assets safe from any type of harm, while cybersecurity is a specific branch focused on defending computer systems and networks from digital attacks.

²³ It is currently not clear in every jurisdiction whether an on-chain transfer is legally dispositive of title, particularly where conflict-of-laws rules, UCC Article 12 (governing controllable electronic records) and other commercial law can have different interpretations. For digital twins, it can be challenging to resolve discrepancies between the on-chain record and the off-chain asset, title, or register. Custody, private-key control, recovery, and operational resilience present related risks, including the possibility that a token transfer is technically valid but not legally or economically effective.

²⁴ For example, the right to erasure under Article 17 of the EU General Data Protection Regulation (GDPR), allows individuals to ask companies and organizations to delete their personal data without undue delay.

assets.²⁵ Market fragmentation can also raise execution and fair pricing concerns. 24/7 trading raises questions about liquidity provision, net asset value (NAV) striking,²⁶ and end-of-day pricing for markets not open 24/7. Atomic settlement can eliminate principal settlement risk by synchronizing delivery and payment, but it may increase intraday liquidity and funding demands by reducing the netting benefits of batch settlement under T+1.

Illiquid or opaque liabilities, such as private credit, are being tokenized to promise instant, 24/7 digital liquidity. Mixing opaque illiquid underlying assets with digital daily liquidity can create structural vulnerabilities. The illiquid asset remains intrinsically illiquid even if its token instrument trades with daily liquidity, creating valuation risks. An asymmetric information problem concerning the underlying assets also remains: Since the off-chain assets often lack observable market prices, valuations will depend on data feeds which may be stale, incomplete, or even manipulated, making it difficult to determine reliable real-time valuations for the on-chain assets. With smart contracts in place, this may even trigger an erroneous transfer, redemption, margin call, or liquidation.

This liquidity mismatch may also leave the financial system more exposed to rapid-fire, systemic market contagion. It has been argued that this might generate more rapid onset of financial crises, including credibility runs and valuation contagion,²⁷ faster execution of fraud by nefarious players, or broader impact of third-party providers being cyber-attacked.²⁸ Cyber security risks will be magnified further by the convergence of tokenization, AI, and quantum computing.²⁹

²⁵ Darrell Duffie. (2026, August 22). Jackson Hole Symposium.

²⁶ Striking a Net Asset Value (NAV) is the official calculation and finalization of an investment fund's per-share price at a specific point in time.

²⁷ Amit Seru. (2024). Preventing Bailouts is Simple, but it Isn't Easy. *Financial Times*. Heightened contagion risk raises the question of whether more circuit breakers will be needed in trading markets to temporarily halt trading to stop panic selling or buying during extreme market drops or spikes.

²⁸ A third-party service provider is an external business or individual hired by a company to perform specific operational functions, services, or tasks.

²⁹ As AI is embedded into tokenized systems, AI-related vulnerabilities will grow meteorically, especially with the prevalence of AI agents executing transactions on the blockchain. Quantum computing is the next amplifying force, but it also represents an existential threat to encryption-based networks, which are therefore transitioning to post-quantum cryptography.

A final obstacle, considered in the next section, is federal regulatory compliance.

Section 4. U.S. Regulation of Tokenized Assets

With the growing interest in digital assets comes the need for regulatory frameworks, providing clarity to implementation and enforcement for those who hold, transfer, or transact in tokenized assets.³⁰ While tokenizing a real-world asset does not, in and of itself, change the nature of the asset, it may change how interests in the asset can be distributed and transferred. At the core of this debate is the question of whether tokenized real-world assets represent a new distribution channel for existing assets, with tokenization in essence providing record-keeping infrastructure that changes format but not substance, or whether digital assets represent new assets. This dichotomy then forms the basis of whether existing securities laws do or do not apply to real-world tokenized assets.

To consider this further, it is helpful to differentiate between digital twin tokenization and natively issued digital assets, and between those created by their issuer and those created by others.³¹ With digital twin tokenization, the digital token represents the real-world asset, while the asset and its legal record, such as title deeds, remain off chain. The token represents ownership rights, but it does not replace the underlying asset. Consequentially, it also does not replace the contractual and regulatory framework. Importantly, this ensures technological neutrality, that is, the digital packaging of an asset does not change its regulatory treatment. Note that the token can be sponsored by the issuer, in which case the holder of the token has ownership, or it can be sponsored by a third party, in which case the token represents synthetic exposure without asset ownership

³⁰ For brevity and broad applicability, I focus here only on U.S. federal, not state, laws.

³¹ This discussion is consistent with the two categories and four models put forth in the (2026, January 28) SEC Staff Statement on Tokenized Securities: Issuer-Sponsored Tokenized Securities that are either issuer-native, such that the token is the on-chain security itself, or issuer-mirrored, so the security is **off chain** and the token conveys no rights but conveys the signal to update it; and Third Party-Sponsored Tokenized Securities that are either custodial, with a claim against a third-party custodian, or synthetic tokenized security, which is a crypto asset representing its own security that provides synthetic exposure to the underlying security.

rights. These cases both contrast with natively issued digital assets created by an issuance event, for which there is no off-chain representation. Importantly, the type of tokenization model impacts the regulatory status of the tokenized asset.

The next step is to determine whether a digital asset is a security under U.S. federal law, regulated by the SEC, or a commodity, regulated by the CFTC. There is currently no comprehensive regulatory framework, creating confusion and unnecessary compliance burdens. The SEC's 2026 crypto interpretive release has narrowed the practical uncertainty by articulating when a tokenized instrument retains the legal status of the underlying security and when a crypto asset falls outside the securities laws. The problem is now therefore less a total absence of guidance than a reliance on a patchwork of regulatory interpretations. A comprehensive statutory framework is still absent, leaving digital asset markets operating across the traditional regulatory lines of securities and commodities and creating continuing confusion and compliance burdens.³²

Legally, a security is generally a tradable financial asset that represents ownership, a debt relationship, or a legal right to share in profits, under the U.S. Securities Act of 1933. It is not, however, always easy to distinguish a security from other types of financial assets or commercial arrangements.³³ Broadly, tokens may be securities when both the underlying asset of a tokenized asset is a security or when the token is offered as part of a security transaction. We recognize that security tokens, which are digital representations of investment products legally classified as securities, such as shares of stock in a company

³² Importantly, this interpretative release does not fully resolve the question. It is not a legislative rule, it is not the product of notice-and-comment rulemaking, so it binds neither the courts nor a future SEC under different leadership, and it may be withdrawn by a successor administration. Therefore, a durable resolution likely requires legislation such as the Clarity Act.

³³ The seminal case is the U.S. Supreme Court decision establishing the "Howey Test" (in SEC v. W.J. Howey Co., 328 U.S. 293 (1946)) to determine whether something not explicitly listed within the Securities Act of 1933 is a "security" within the SEC's jurisdiction. According to the Howey Test, an investment contract (which is, among other things, one of the assets enumerated in the definition of "security" in each of the federal securities laws) generally involves an investment of money in a common enterprise, with an expectation of profit derived from the efforts of others.

or claims to private equity, are also securities. Consequentially, the SEC classifies tokenized equities the same under federal law as it does regular equity securities.³⁴

On March 11, 2026, the SEC and CFTC entered into a memorandum of understanding to reduce jurisdictional overlap and forum uncertainty for digital assets that potentially straddle the securities-commodities line. The framework classifies digital assets according to their characteristics, uses, and functions. There are digital commodities, such as Bitcoin, Ether, and similar assets whose value does not depend on the managerial efforts of a promoter, which fall under CFTC jurisdiction. There are digital collectibles, which are non-fungible items held for consumption or collectible value. There are digital tools, which are tokens that provide access or utility within a network rather than an investment return. There are stablecoins, which are payment instruments regulated under the GENIUS Act framework. Finally, there are digital securities, which are instruments representing ownership, debt, or profit participation and fall under SEC jurisdiction. The first three categories are generally not securities under the federal securities laws by classification. Importantly, however, classification governs the asset, not the transaction: a non-security digital asset can still be sold pursuant to an investment contract, and that transaction can require registration under the Securities Act of 1933 even though the asset itself is not a security. Because the framework is a memorandum of understanding between the agencies, it allocates enforcement and interpretative priorities rather than changing the underlying statutes.

The Digital Asset Market Clarity Act (Clarity Act), which was approved by the U.S. House of Representatives in July 2025 but failed to pass a procedural Senate vote in September 2026, seeks to establish market structure rules for blockchain networks and to

³⁴ If an instrument is a security, any public offering must be registered with the SEC or qualify for an exemption, with the related disclosure obligations and potential liability under Sections 11 and 12 of the Securities Act. Persons effecting transactions in the asset generally must register as broker-dealers under the Exchange Act and comply with applicable FINRA, custody, and net capital rules. Persons providing advice about the asset may need to register as investment advisers and owe fiduciary duties. A pooled vehicle holding the asset may be an investment company subject to the Investment Company Act or must fit an exemption. The antifraud and insider-trading prohibitions of Section 10(b) and Rule 10b-5 apply, together with Regulation FD, beneficial-ownership reporting, and Section 16 short-swing rules for registered issuers. Trading venues may also need to register as national securities exchanges or alternative trading systems.

comprehensively define regulatory responsibilities for the SEC and CFTC across digital asset markets. The SEC abruptly postponed its vote on “Regulation Crypto Assets,” a tokenization exemption in light of the stalled legislative talks over Section 10505 of the Clarity Act, governing digital asset market structure, which would establish the safe harbor that tokenized traditional instruments retain their legal status as securities.³⁵ Under the Clarity Act, the SEC would retain oversight of digitized securities, and the CFTC would have regulatory primacy over digital commodities. This would affirm that placing traditional financial instruments or shares on chain does not alter their baseline definition or regulation as federal securities, which would be an important step forward in clarifying regulatory oversight. SEC Chairman Atkins made clear his intention, “with or without that legislation,” to proceed with the agency’s crypto rulemaking agenda “for how crypto assets are issued, traded, transferred and held.”³⁶

The regulatory environment remains dynamic. Market participants need to monitor the rapidly evolving legislative and regulatory landscape, as the future path of tokenized assets will likely depend in part on the regulatory decisions to be made in the next months and years.

Section 5. Conclusion

The nature of how a tokenized real-world asset is issued, held, traded, and settled is changing rapidly and has the potential to be transformative for financial markets. Because we are at the inception of these nascent assets, it is vital to identify guiding regulatory principles for policymakers.

One key principle, which I highlighted at the start of Section 4, is technological neutrality. It focuses on the underlying economics of an asset, recognizing that putting an asset on a blockchain changes its technological wrapper but does not necessarily change

³⁵ “Regulation Crypto Assets,” proposed August 18, 2026, includes two exemptions from the registration requirements of the Securities Act of 1933 specifically tailored to certain investment contracts involving crypto assets.

³⁶ Remarks at the Solana Policy Institute Summit: Washington X Wall Street, September 14, 2026.

its substantive underlying economics. Assets should be subject to consistent technology-neutral regulation, commensurate with the risks they pose.³⁷

We have the unique opportunity now, at the nascent stages of digital real-world asset adoption, to take those steps necessary to harness tokenization’s advantages, while mitigating its risks, to build a smarter, safer, more broadly accessible, flexible, and efficient market system.

³⁷ A related principle, which is summarized by the Financial Stability Board in its global regulatory framework for crypto-asset activities in 2023 as “same activity, same risk, same regulation,” is functional equivalence. For further discussion specifically related to blockchain-based systems, see De Fillippi, P. and Mannan, M. (2025), “Regulatory Equivalence in Blockchain Systems: The Role of Public Values and Legitimacy,” in J. Goosens, E. Keymolen, & A. Stanojevic (eds), *Public Governance and Emerging Technologies: Values, Trust, and Regulatory Compliance*, 1 edition, Springer, pp. 151-177.

Appendix. Glossary

Cryptocurrencies: Cryptocurrencies are digital assets that use cryptography and a distributed ledger to secure and record transactions, typically without a central issuer or administrator.

Distributed Ledger: A distributed ledger is a cryptographically secure digitally distributed decentralized ledger recording transactions shared among network participants, ensuring that changes are agreed upon by all parties rather than subject to a single point of control or failure.

Blockchain: Blockchain technology is a shared, immutable digital ledger that enables the recording of transactions and the tracking of assets. Blockchain is also specifically the name of the distributed technology ledger underlying Bitcoin, the world's first decentralized digital currency.

Digital native: A digital native token is issued directly on its home blockchain.

Digital twin: A digital twin ("wrapped" token) is an on-chain representation linked to an off-chain asset or legal record; the token's rights depend on the legal and operational mechanism that reconciles the two.

Off Chain: Off chain refers to an asset or activity taking place on non-tokenized (conventional) systems.

On Chain: On chain refers to an asset or activity being operated on a programmable ledger.

Permissioned Ledger: A permissioned ledger is a distributed ledger in which participation, transaction validation, or access to data is restricted to approved parties.

Pseudonymous: Pseudonymity acts as a layer of separation between your real-world identity and your online activities. It replaces your legal name with a consistent digital identifier or "fictional persona", and it is typically your wallet address in cryptocurrency.

Smart Contract: A smart contract is a self-executing computer program stored on a blockchain that automatically carries out actions when preset rules and conditions are met, such as defining the rules for asset ownership and transfer.

Stablecoins: A stablecoin is a type of cryptocurrency designed to maintain a fixed value by tying, or "pegging," it to a reserve asset like the U.S. dollar.

Token: A token is generally a smart-contract-based asset created on a distributed ledger network.

Tokenized real-world assets: Tokenized real-world assets are smart-contract-based assets issued on a shared distributed ledger, as defined in World Economic Forum, “Asset Tokenization in Financial Markets: The Next Generation of Value Exchange,” May 2025.